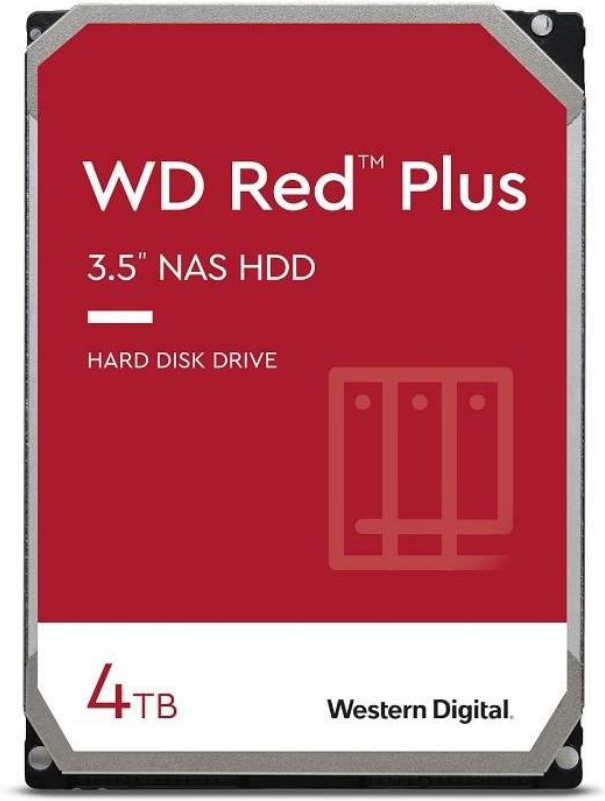
HARD DRIVE

HDDs store data on spinning magnetic disks coated with magnetic material. They use read/write heads to access and modify data on these disks.

Advantages: Generally cheaper for larger storage capacities. Commonly used in PCs and data centers.

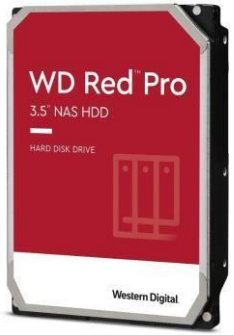
Disadvantages: Slower read/write speeds compared to SSDs. Prone to mechanical failure due to moving parts.





TYPE OF HARD DRIVE

Hard Disk Drive (HDD)



•**Advantages:** Generally cheaper for larger storage capacities. Commonly used in PCs and data centers.
•**Disadvantages:** Slower read/write speeds compared to SSDs. Prone to mechanical failure due to moving parts.

Solid State Drive (SSD)



•**Advantages:** Faster read/write speeds, more resistant to physical shock, and more energy-efficient. Ideal for laptops, desktops, and high-performance computing.
•**Disadvantages:** Generally more expensive per GB compared to HDDs.

Hybrid Drives (SSHD)



•**Advantages:** Offers a balance between speed and storage capacity by using SSD-like performance for frequently accessed data.
•**Disadvantages:** Not as fast as pure SSDs for all data access, and they're still susceptible to mechanical failure like traditional HDDs.

External Hard Drives



•**Advantages:** Portable, easy to connect to multiple devices, and offer additional storage capacity.
•**Disadvantages:** Speed might be limited by the connection interface (e.g., USB 2.0 vs. USB 3.0/3.1).

Enterprise-Class HDDs and SSDs



•**Advantages:** Designed for heavy workloads, high data transfer rates, and increased reliability.
•**Disadvantages:** Usually more expensive compared to consumer-grade drives.

NAS (Network Attached Storage) Drives



•**Advantages:** Designed for 24/7 use, optimized for RAID configurations, and typically have higher durability.
•**Disadvantages:** May have lower rotational speeds or higher latency compared to performance-focused drives.

ABOUT RAID

RAID (Redundant Array of Inexpensive Disks) is a data storage virtualization technology that combines multiple disk drives into a single logical unit for faster performance, better hardware failover, and improved disk Input/Output reliability. Let's go over some common questions regarding RAID and see what the best RAID configuration is for your server.

Pro performance

Get the performance and protection you need for data-intensive creative workflows.

Production power

RAID solutions can be configured for the most rigorous uses, such as time-sensitive, data-sensitive post-production workflows.

Cloud-friendly

Cloud-ready RAID storage can be ideal for both business and home users that want to access their data from anywhere.

Fault tolerant

RAID can deliver high data protection and redundancy whenever uptime and availability are critical needs.



RAID CONTROLLER

A **RAID Controller** is a device used to manage hard disk drives in a storage array. It can be used as a level of abstraction between the OS and the physical disks, presenting groups of disks as logical units. Using a RAID controller can improve performance and help protect data in case of a crash.



Hardware vs Software RAID

There are two types of RAID, each with their advantages and disadvantages.

Hardware RAID: With a hardware RAID setup, the drives connect to a RAID controller card inserted in the server's motherboard. This provides optimal performance, as the processing is handled by the RAID card rather than the server, providing less strain when writing backups and restoring data.

Software RAID: When drives are connected to the server motherboard without a RAID controller, the configuration is managed by the Operating System's software. This is the cheaper option; all you need to do is connect the drives and configure the OS. However, you are restricted to the RAID levels your OS can support.

Hardware RAID provides more flexibility via more configuration options and frees you from the limitations of a software RAID. Depending on the complexity of your RAID configuration, performance could be an issue if you are using software RAID.



What RAID Configuration is Best?

Not all RAID configurations are created equal in terms of redundancy, speed, or disk size.

Some common RAID levels include RAID 0, RAID 1, RAID 5, RAID 10. Here is a very simple TLDR chart. Read on for a more detailed version of the pros and cons of each.

Raid Level	Minimum Disks	Performance	Redudancy
RAID 0	2	Excellent	None
RAID 1	2	Good	Excellent
RAID 5	3	Good	Good
RAID 1+0	4	Excellent	Excellent



BENEFITS OF RAID

- Improved cost-effectiveness because lower-priced disks are used in large numbers.
- Using multiple hard drives enables RAID to improve the performance of a single hard drive.
- Increased computer speed and reliability after a crash, depending on the configuration.
- Reads and writes can be performed faster than with a single drive with RAID 0. This is because a file system is split up and distributed across drives that work together on the same file.
- There is increased availability and resiliency with RAID 5. With mirroring, two drives can contain the same data, ensuring one will continue to work if the other fails.



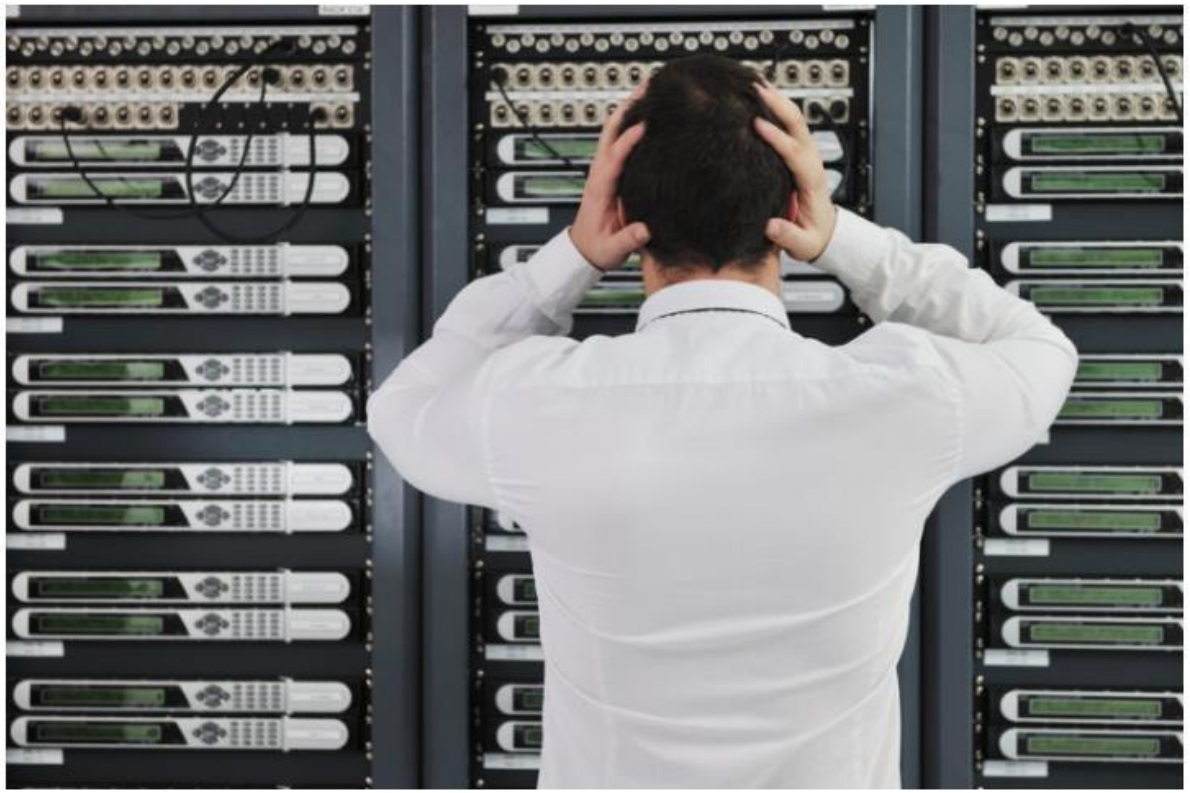
DOWNSIDES OF USING RAID

- Nested RAID levels are more expensive to implement than traditional RAID levels, because they require more disks.
- The cost per gigabyte for storage devices is higher for nested RAID because many of the drives are used for redundancy.
- When a drive fails, the probability that another drive in the array will also soon fail rises, which would likely result in data loss. This is because all the drives in a RAID array are installed at the same time, so all the drives are subject to the same amount of wear.
- Some RAID levels -- such as RAID 1 and 5 -- can only sustain a single drive failure.
- RAID arrays, and the data in them, are vulnerable until a failed drive is replaced and the new disk is populated with data.
- Because drives have much greater capacity now than when RAID was first implemented, it takes a lot longer to rebuild failed drives.



DOWNSIDES OF USING RAID

- If a disk failure occurs, there is a chance the remaining disks may contain bad sectors or unreadable data, which may make it impossible to fully rebuild the array.



RAID 0

•**RAID 0** stripes data at the block level across each of the disks in the array. Because the data is striped across multiple disks, you will see improved read and write performance, especially for larger files. This is due to the ability of the array to simultaneously read or write a block to each of the disks in the array. As such, the performance will tend to scale with the number of disks within the array. A RAID 0 can contain two or more disks, with the maximum number being limited based on the number of disks supported by your RAID controller and available drive bays for your server. Because RAID 0 spreads storage blocks across all of the disks in the array, any disk failure will result in the entire array being lost. As such, the risk of data loss due to a drive fault will also scale with the number of disks within the array. We do not recommend using a RAID 0 for production systems.

•Capacity: $VD = (C * n) \mid \text{VirtualDisk} = (\text{Capacity of one disk} * \text{number of disks})$

•I/O Performance: Greatly improved read and write performance

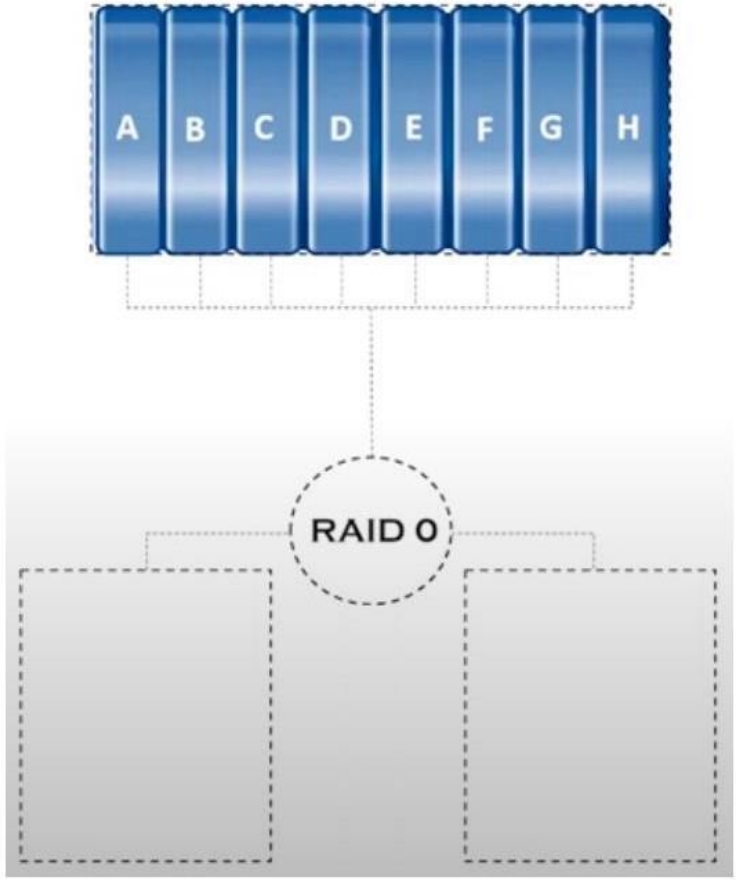
•Fault Tolerance: None, increased risk of failure that scales with number of disks in the array



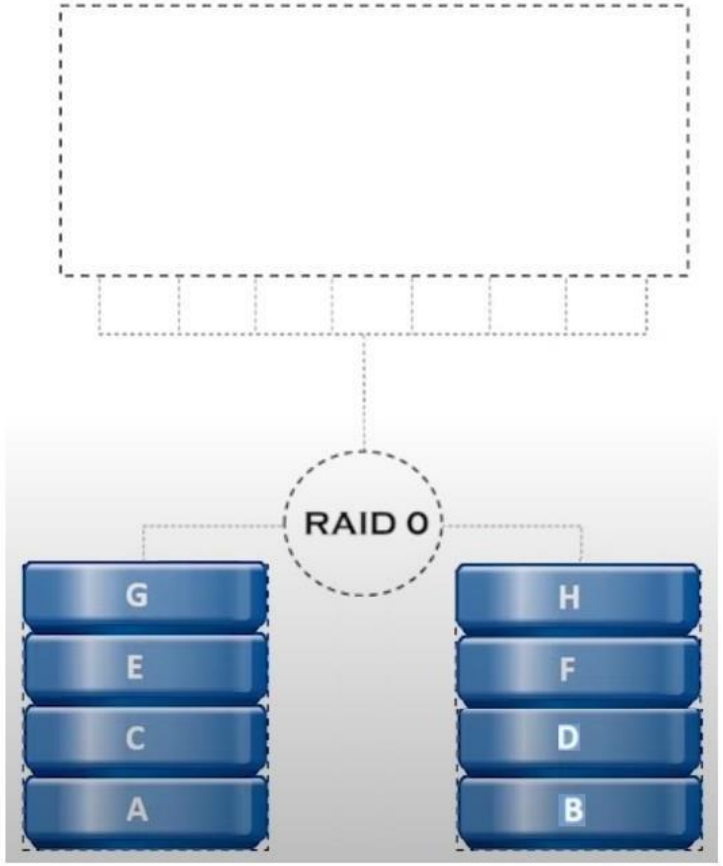
RAID 0



BEFORE



AFTER





RAID 1

•**RAID 1** mirrors data at the block level on two disks. Each time a data block is written to the storage subsystem, your RAID controller will write that same data block on both disks, providing fault tolerance should one of the disks fail. Because of this, the I/O performance characteristics will have slightly lower write performance compared to a single disk, while read performance will be nearly double that of a single disk. This is because when the RAID controller needs to write data it has to do so on both disks, however, when it needs to read data it can simultaneously fetch two different reads from either disk since the data set on both disks is mirrored. Because the data is mirrored on both disks in the array, either one of the disks can fail, as log as one remains functional, without causing the array to fail. NOTE: While most hardware based RAID controllers typically only support two disks in a RAID 1 array, some controllers and software RAID systems may support more than two disks. In these cases, write performance will slightly degrade as the number of disks increase and the read performance will scale with the number of disks.



•Capacity: VD = (C * n) / n | VirtualDrive = (Capacity of one disk * number of disks) / number of disks

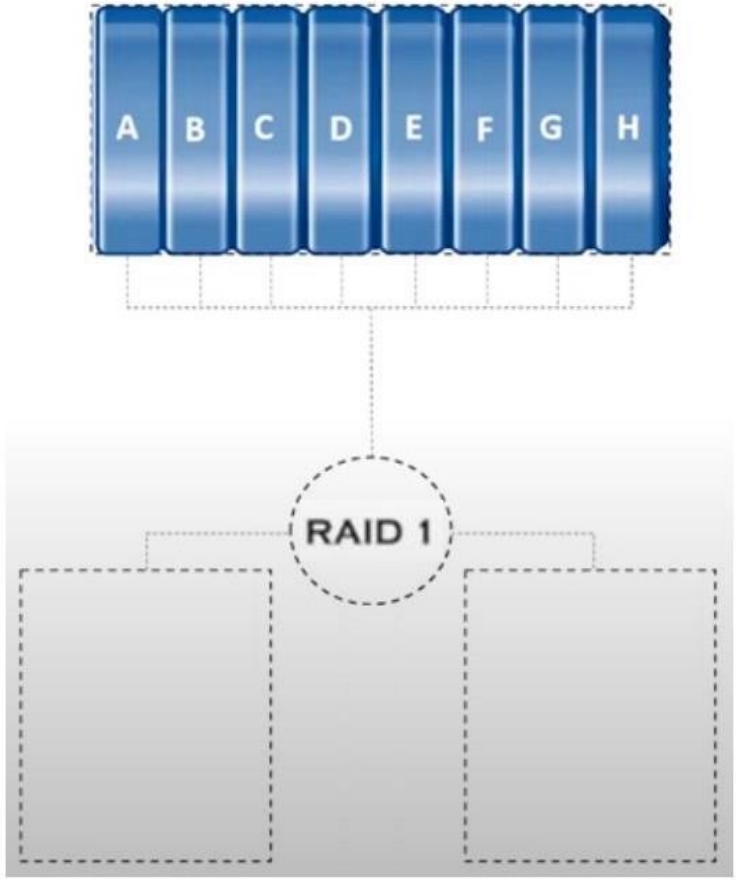
•I/O Performance: Slightly lower write performance with nearly double the read performance

•Fault Tolerance: n / n

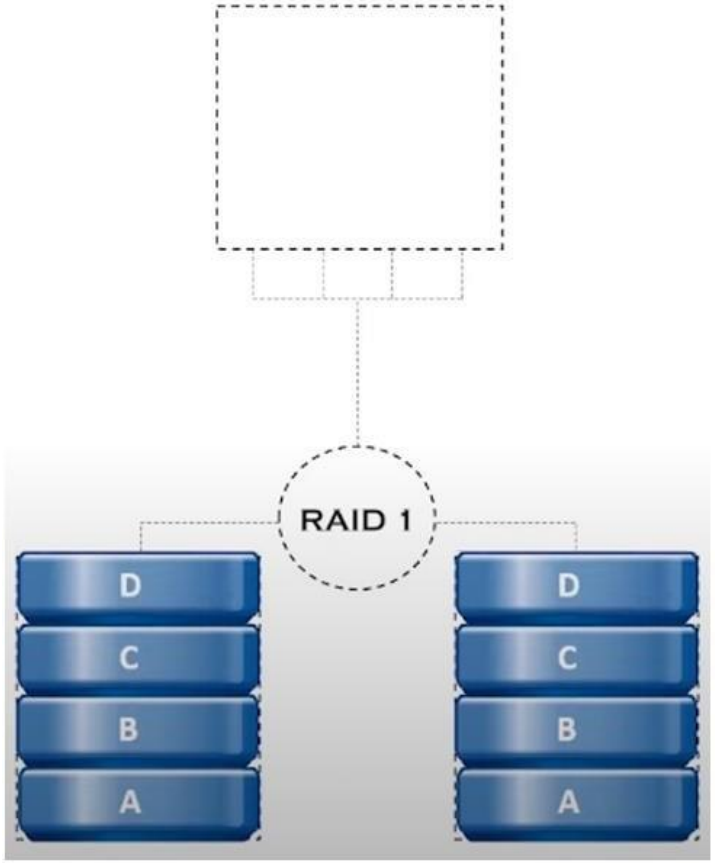
RAID 1



BEFORE



AFTER





RAID 5

•**RAID 5** stripes data at the block level across each of the disks in the array, while also calculating parity data, which is distributed across the disks in the array to add fault tolerance. With a RAID 5 array, there will be one parity set distributed across the array, which allows for any one disk in the array to fail without causing the array to fail. Because the distributed parity needs to contain parity information for one disks worth of information, the raw space provided by the array will be the total capacity of all of the disks minus one disks worth of capacity. This is also why a RAID 5 has a minimum requirement of three disks. The I/O performance for RAID 5 will offer improved read and write performance, however, writes will not scale as linearly as a RAID 0 due to the RAID controller needing to perform the parity calculation and also store the resulting parity data. NOTE: While some software RAID systems support RAID 5, we do not recommend this in a production because all of the parity calculations will need to be done by your servers CPU rather than the dedicated specialized hardware controller that is designed to accelerate these types of calculations. As such, a software RAID 5 will see CPU load scale with disk I/O and can often cause overloads as disk writes increase.



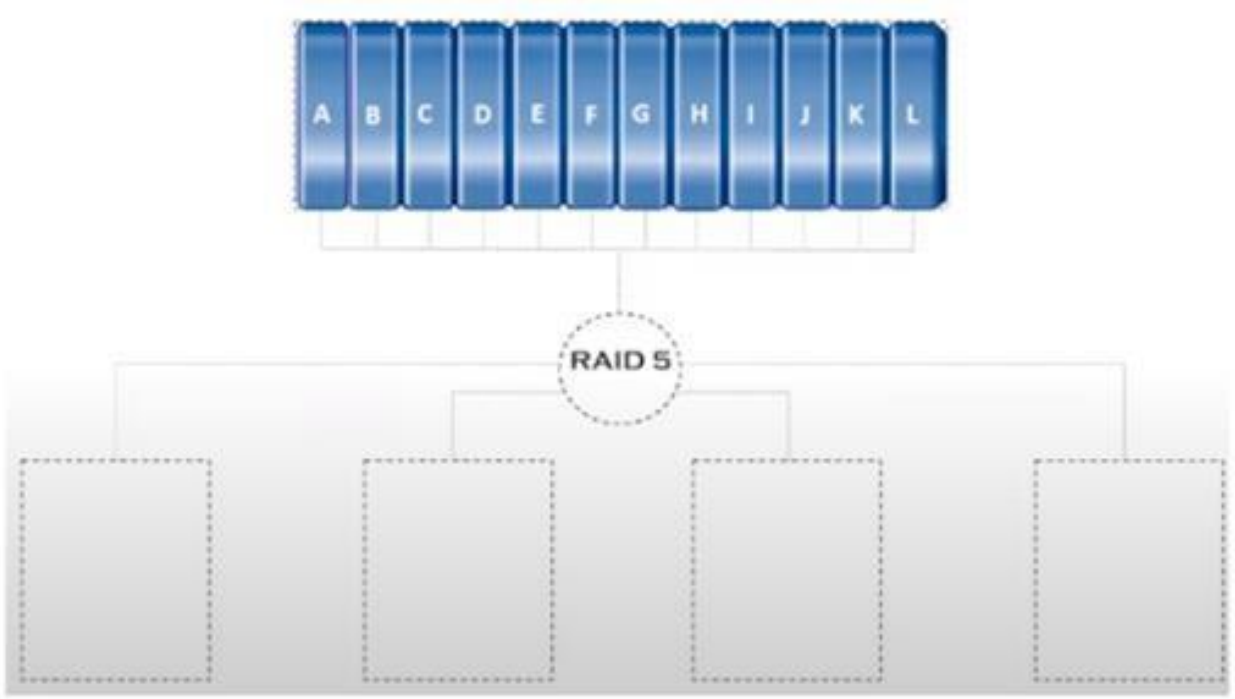
•Capacity: $VD = (C * n) - C$ | VirtualDrive=(Capacity of one disk * number of disks) – Capacity of one disk

•I/O Performance: Improved read and write performance

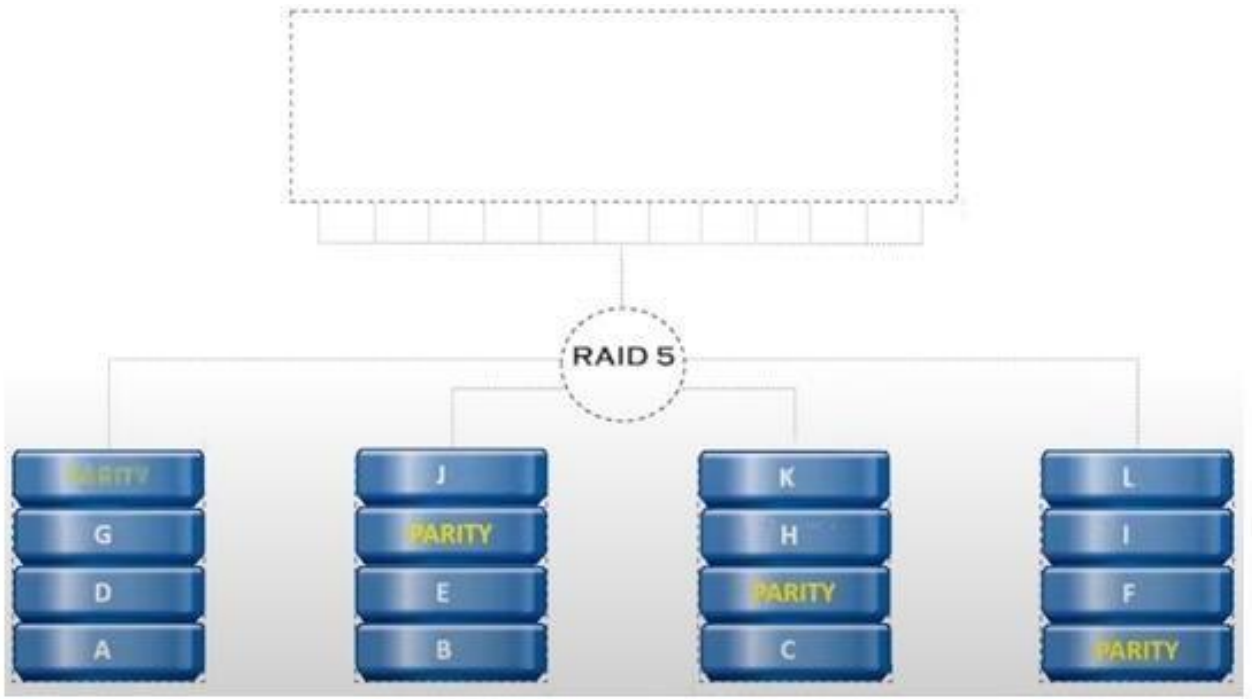
•Fault Tolerance: $n - 1$

RAID 5

BEFORE



AFTER



RAID 6

•**RAID 6** This technique is similar to RAID 5, but it includes a second parity scheme distributed across the drives in the array. The use of additional parity enables the array to continue functioning, even if two disks fail simultaneously. However, this extra protection comes at a cost. RAID 6 arrays often have slower write performance than RAID 5 arrays.

•Capacity: VD = (C * n) – C | VirtualDrive=(Capacity of one disk * number of disks) – Capacity of one disk

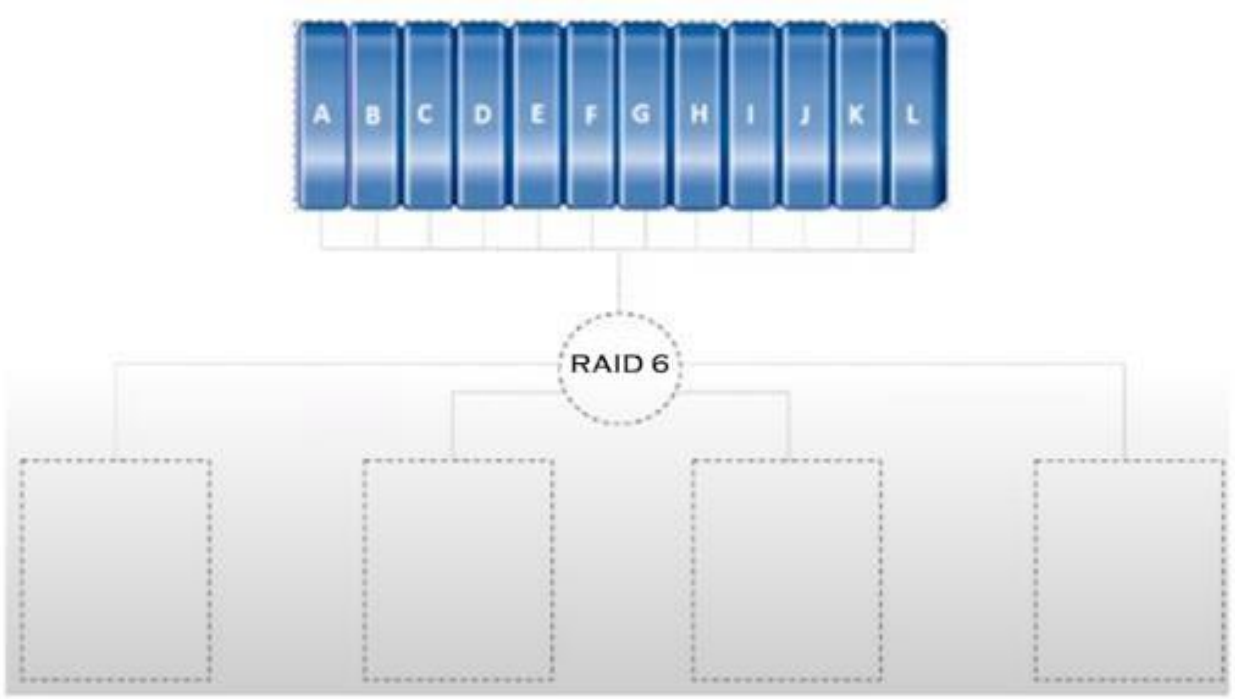
•I/O Performance: Improved read and write performance

•Fault Tolerance: n – 1

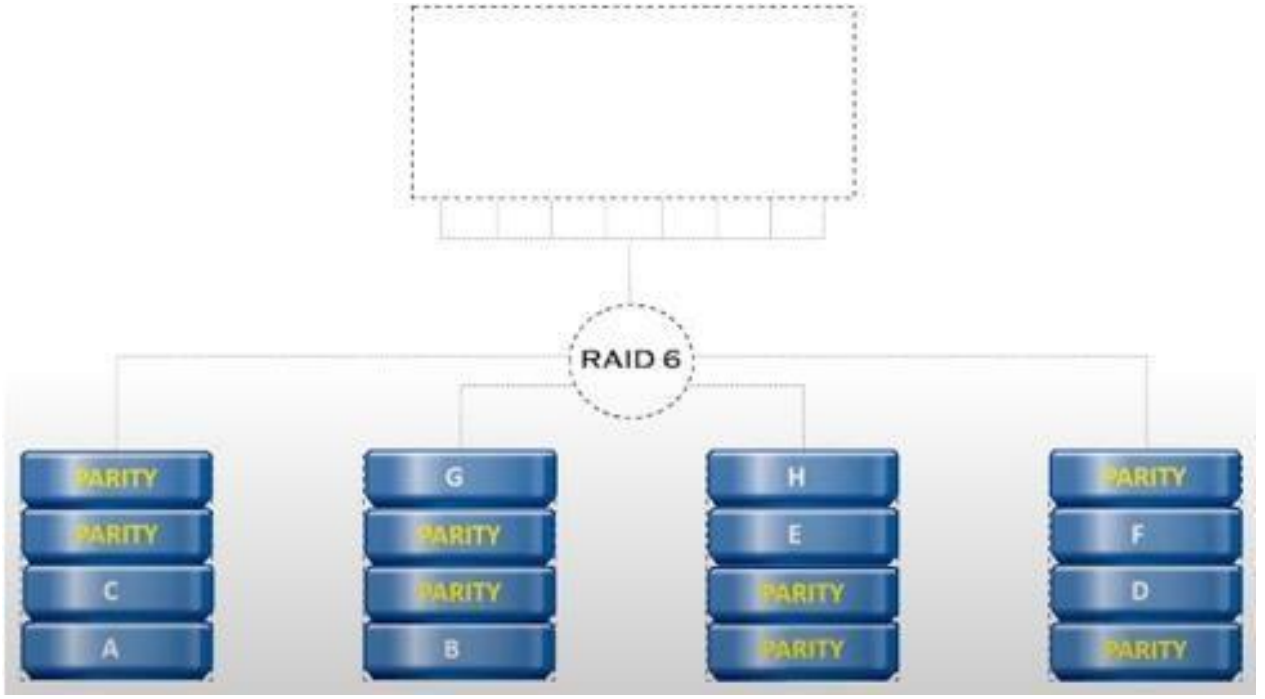


RAID 6

BEFORE



AFTER



RAID 10

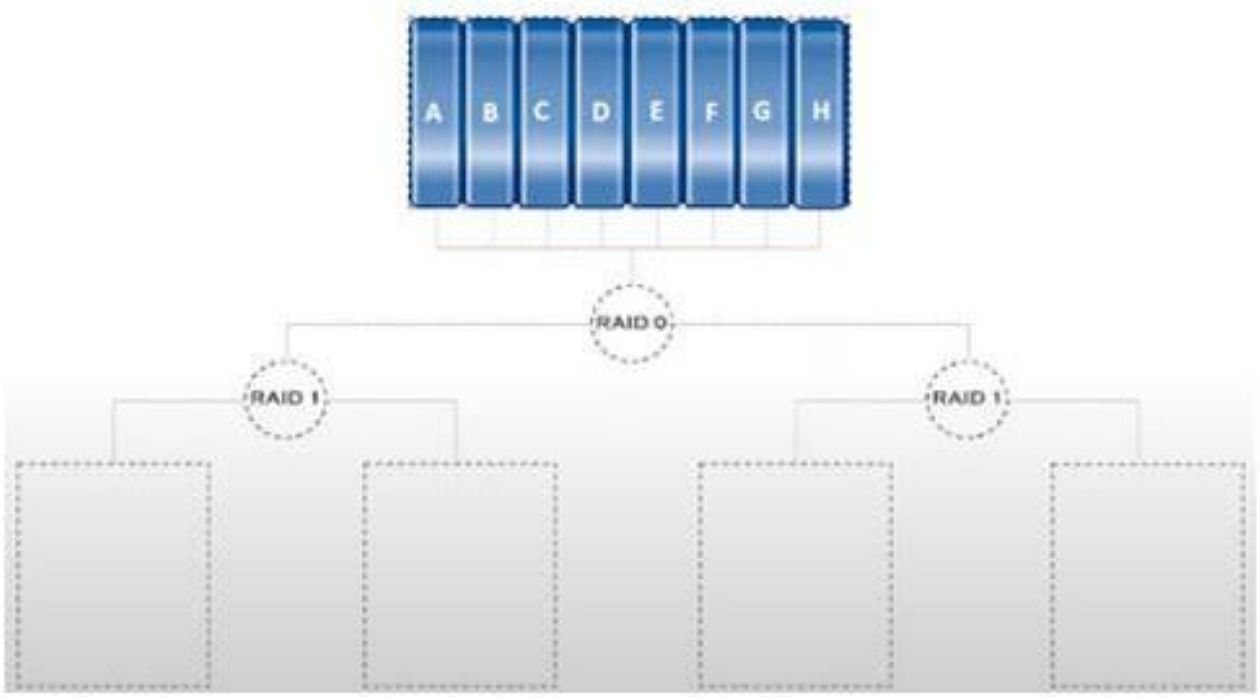
•**RAID 10** is a nested array containing RAID 0 of RAID 1 sets; in other words it is a RAID array of multiple RAID arrays. As such, this RAID level combines some of the performance characteristics of both RAID 0 and RAID 1. The I/O performance will be improved for both read and writes, which will scale differently based on the number of disks in the array. The read performance will scale with the number of disks in the array, while the write performance will scale with the number of RAID 1 sets in the array. For example, a RAID 10 with four disks will have two RAID 1 sets that are then part of a RAID 0, therefore the read performance will be nearly four times a single disk in the array, and write performance will be nearly twice that of a single disk. This RAID level offers a great combination of performance and fault tolerance. Half of the disks in the array can fail without the array failing. This is because one disk in each of the RAID 1 sets can fail without that RAID set failing. Because of these characteristics, you will notice that the minimum number of disks required will be four and the total must be an even number of disks, such that increasing the number of disks in the array is done in pairs of disks. While the number of disks required compared to the raw useable capacity is lower, this RAID level offers the best overall combination of performance and fault tolerance. This is a great option for read heavy databases or mixed work loads.



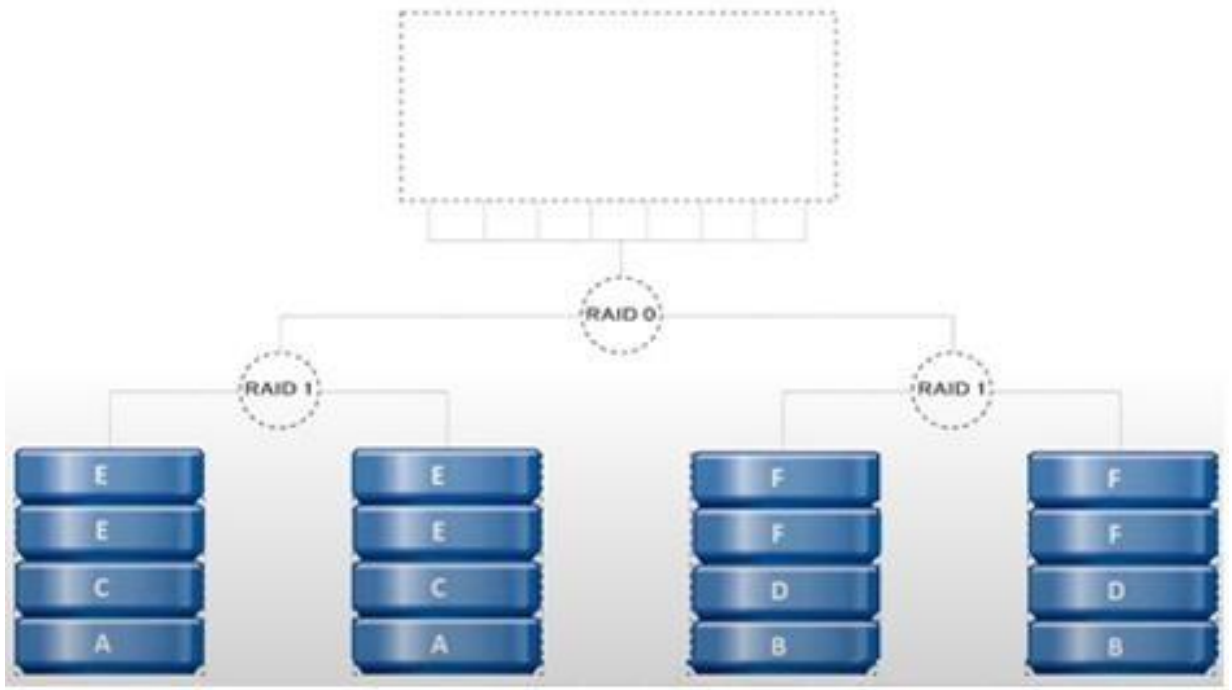
•Capacity: VD = (C * n) / n | VirtualDrive = (Capacity of one disk * number of disks) / number of disks
•I/O Performance: Improved write performance with greatly improved read performance
•Fault Tolerance: n / 2

RAID 10

BEFORE



AFTER



Interview Q&A



1. What is RAID and why is it used?
Answer: RAID is a technology that combines multiple disk drives into a single logical unit. It is used primarily for improving data reliability, increasing performance, or both, depending on the RAID level implemented.

2.What are the different RAID levels, and how do they differ?
Answer: There are several RAID levels:
•**RAID 0:** Stripes data across multiple drives for improved performance but offers no redundancy.
•**RAID 1:** Mirrors data across two drives to provide redundancy without performance improvement.
•**RAID 5:** Distributes parity across multiple drives for both performance and fault tolerance.
•**RAID 6:** Similar to RAID 5 but with dual parity for increased fault tolerance.
•**RAID 10 (or RAID 1+0):** Combines mirroring and striping to offer both redundancy and performance benefits.

3. What is parity in RAID, and how does it help with fault tolerance?
Answer: Parity is used to calculate and store redundant information across multiple drives in RAID arrays. In case of a drive failure, the parity information can be used to reconstruct the lost data, ensuring fault tolerance.

4. What are the advantages and disadvantages of RAID 0?
Answer: Advantages: Improved performance due to data striping across multiple drives. **Disadvantages:** No redundancy, so if one drive fails, all data is lost.

5. Explain the difference between hardware RAID and software RAID.
Answer: Hardware RAID: Relies on a dedicated RAID controller to manage the array independently of the host system's CPU. It often provides better performance but can be more expensive.
Software RAID: Relies on the host system's CPU for managing the RAID array. It's usually more cost-effective but might not offer the same level of performance as hardware RAID.

6. How does RAID 5 recover data in the event of a drive failure?
Answer: RAID 5 uses distributed parity across multiple drives. If one drive fails, the parity information along with the data from the remaining drives can be used to reconstruct the lost data.

7. What is a hot spare in RAID?
Answer: A hot spare is an unused drive designated to automatically replace a failed drive in a RAID array. When a failure occurs, the hot spare is automatically brought online to start the rebuild process without manual intervention.

8. Can you add drives to a RAID array? If so, how?
Answer: Yes, it's possible to add drives to certain RAID levels (like RAID 5 or RAID 6) to expand the array's capacity. This often involves using a RAID management tool to initiate the expansion process, which typically requires redundancy to be maintained during the expansion.

9. What considerations should be made when choosing a RAID level for a particular system?
Answer: Consider factors like performance requirements, desired level of fault tolerance, cost, and available drive capacity when choosing a RAID level for a system.

10. What is RAID striping and how does it improve performance?
Answer: RAID striping is the process of writing data across multiple drives in small, evenly-sized chunks. This improves performance by allowing multiple drives to work concurrently to read and write data, increasing throughput.

11.What Are The Different States An Array Can Be In And Explain Each State?
Answer: Online, Downgraded, Offline, Rebuilding.

Interview Q&A



12.Explain Online, offline, degraded States Of An Array?
Answer: Online: when all drives are working fine.
Downgraded: Whenever there is a drive failure but still the array is functioning fine
Offline: Array or whole data storage is down
Rebuilding: Storage access is there but since a new drive has been inserted in place of a failed drive data is being written to new drive which might slow down the performance of the whole RAID array.

13. What Is The Difference Between A Global Hot Spare & A Dedicated Hot Spare?
Answer: Global hotplate is available for the any array in the whole enclosure or Storage subsystem.
If there is an enclosure having 10 drives & we have 3 drives in RAID5(1st array) , 3 more drives in second RAID5(2nd array) & 2 more drives in RAID 1 config. We can specify in RAID config utility whether a dedicated hot spare is assigned for 1st RAID5 array. If there is a drive failure in 2nd or 3rd array this dedicated hot spare will not be involved there. But if the array for which this is dedicated has any drive failure this dedicated hot spare takes over.

14. How Raid Is Configured Through Bios?
Answer: If we have a Hardware RAID controller card it gives an option while machine booting to enter into RAID BIOS utility. Here we have options which give us options to create RAID using a semi-GUI(DOS based GUI) interface.

15. How Raid Is Configured In OS Level?
Answer: Once we install device drivers & also RAID config or management utility using that we can configure RAID in OS level.

16.What Is The Difference Between A Software Raid VS Hardware Raid?
Answer: See page 4.

17. What Is A Hot Spare?
Answer: Hot spare is an extra, unused disk drive that is part of the disk subsystem. It is usually in standby mode ready for service if a drive fails. Whenever there is a drive failure this hot spare kicks in and takes over that failed drive's role.

18. When Jbod (Just A Bunch Of Disks) Is Preferred Over Raid Array?
Answer: When there is no need for redundancy & when it is ok if there is some hard disk failure or data unavailability in such scenarios JBOD is preferred over RAID because JBOD is inexpensive storage solution. It is also easy to setup & start using compared to RAID.

19. What is the Difference between a Jbod & a Raid Array?
Answer: Hard disks that are not configured in a RAID configuration. They are just disks piled or connected in one single enclosure.
RAID is having the advantage of bearing a disk failure & still give data availability.

20. What Is Check Consistency?
Answer: Consistency check or CC verifies correctness of data in logical drives. This is a feature of some of the RAID hardware controller cards.

21. What Is Initialization?
Answer: Initialization is the process of preparing a drive for storage use. It erases all data on the drive and makes way for new file system creation.